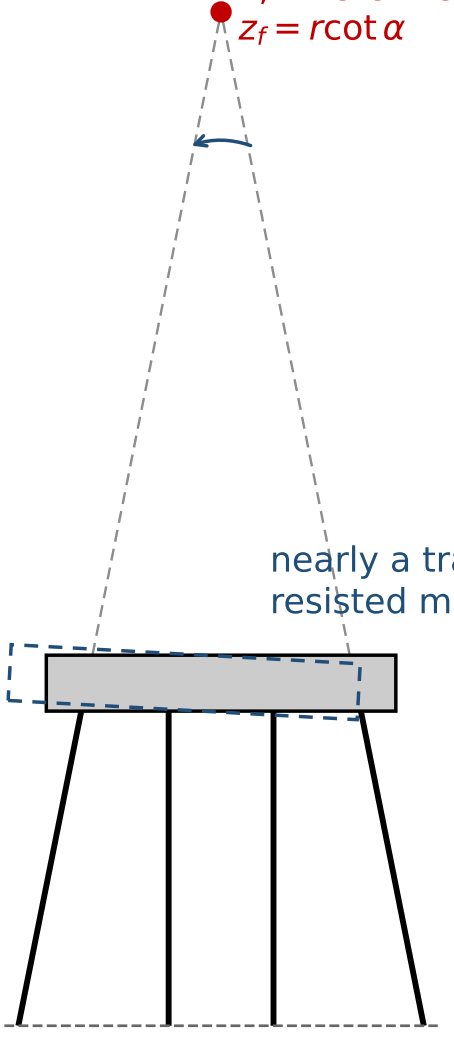
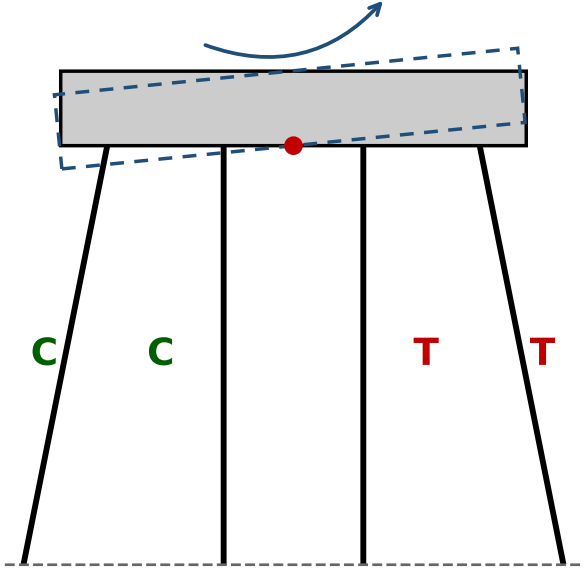
(a)

F, where the pile axes meet

$z_f = r\cot\alpha$

nearly a translation at the cap,
resisted mainly by pile bending



(b)

rotation φ about a centre near the pile heads,
resisted by the axial pile stiffness